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Annual Review of Animal Biosciences

Rumen-Targeted Mining of Enzymes for Bioenergy Production

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Annu. Rev. Anim. Biosci. 2025. 13:343–69

First published as a Review in Advance on November 14, 2024

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-021022-030040

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Keywords

bioenergy, biofuels, rumen, enzymes, lignocellulose

Abstract

Second-generation biofuel production, which aims to convert lignocellulose to liquid transportation fuels, could be transformative in worldwide energy portfolios. A bottleneck impeding its large-scale deployment is conversion of the target polysaccharides in lignocellulose to their unit sugars for microbial fermentation to the desired fuels. Cellulose and hemicellulose, the two major polysaccharides in lignocellulose, are complex in nature, and their interactions with pectin and lignin further increase their recalcitrance to depolymerization. This review focuses on the intricate linkages present in the feedstocks of interest and examines the potential of the enzymes evolved by microbes, in the microbe/ruminant symbiotic relationship, to

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depolymerize the target polysaccharides. We further provide insights to how a rational and more efficient assembly of rumen microbial enzymes can be reconstituted for lignocellulose degradation. We conclude by expounding on how gains in this area can impact the sustainability of both animal agriculture and the energy sector.

INTRODUCTION

The process of photosynthesis enables plants to harvest energy from the sun, storing it mostly in polymeric compounds known collectively as lignocellulose in the plant cell wall. Lignocellulose, a macromolecular complex, also represents the most abundant biomass on our planet and is composed mainly of two carbohydrate polymers, cellulose and hemicellulose, and a noncarbohydrate polymer lignin (1, 2) (**Figure 1**). By its sheer volume and availability, lignocellulose constitutes the most attractive source of feedstock for bioenergy production. Thus, in the field of second-generation bioenergy production, the aim is to depolymerize its carbohydrate polymers to their unit sugars for fermentation by microbes to liquid fuels (3) such as ethanol and butanol.

In first-generation biofuel production, microorganisms are harnessed to ferment sugars released from less recalcitrant feedstocks, including sucrose from sugarcane and starch from corn (4).

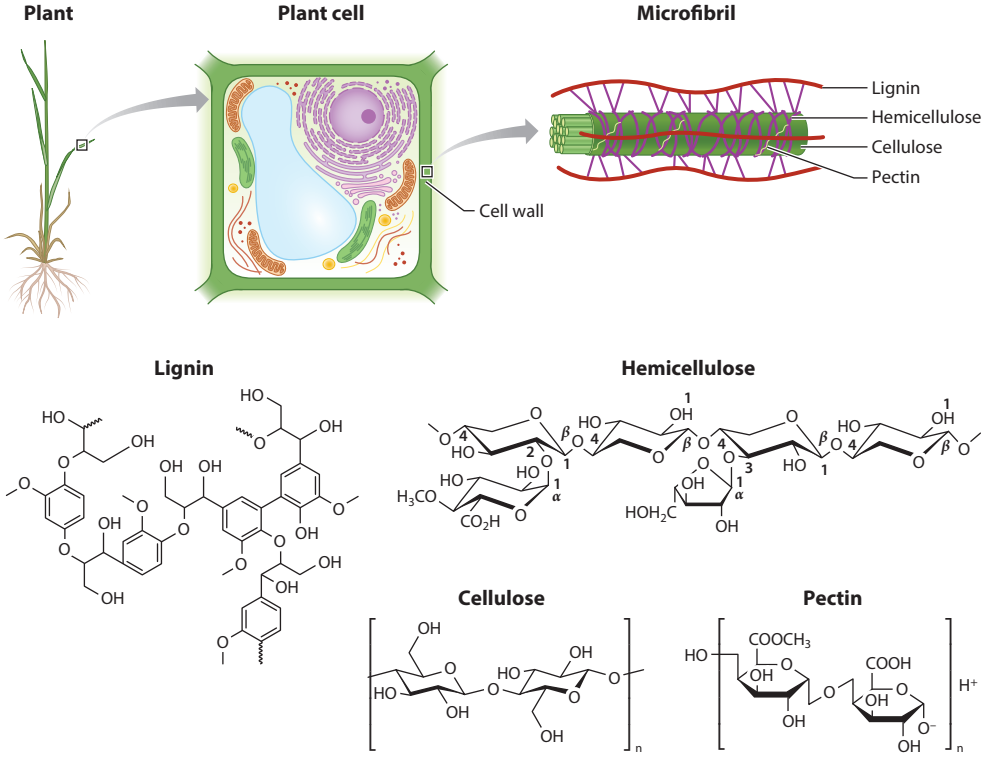

Figure 1

The plant cell wall and its components. An illustration of the plant; the plant cell wall; and the arrangement of cellulose microfibrils encased in a complex maze of hemicellulose, pectin, and lignin. The primary linkages present in these major plant cell wall components are also presented.

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This process requires only a few enzymes to release the fermentable sugars from the feedstock and is a mature technology currently carried out at industrial scale without significant bottlenecks (1). The use of feedstocks such as corn, wheat, and soybean, which are seen as food sources for an ever-expanding human population, often poses ethical concerns for the field of bioenergy production. This dilemma associated with feedstocks of first-generation biofuels thus makes lignocellulose an even more attractive feedstock. However, unlike hydrolysis of sucrose and starch, unlocking the sugars in lignocellulose continues to pose a challenge and represents a significant constraint to the second-generation biofuel industry (1).

We examine the current feedstocks targeted for second-generation bioenergy production, their intricate structures, and the enzymes evolved in the microbial world, especially in the ruminant first stomach or rumen, to cleave the complex array of linkages to yield unit sugars for fermentation. We further present the niche of major plant cell wall–degrading organisms in the rumen, a better understanding of the roles of which can lead to a rational assembly of more efficient lignocellulose-degrading enzyme cocktails. Furthermore, we present insights on how new enzymes may be prospected to augment the rationally reconstituted enzyme cocktails. We expect that the new insights provided in this review will rekindle the interest and efforts targeting the rumen microbes to harness their enzymes evolved over millennia to degrade and ferment lignocellulose. In addition to augmenting our knowledge on rumen plant cell wall metabolism and how it may be improved to ameliorate its impact on the environment, harnessing these enzymes to aid advancement of second-generation bioenergy production could reduce reliance on fossil fuels.

PLANT CELL WALL COMPONENTS SIGNIFICANT TO BIOENERGY PRODUCTION

Cellulose

Cellulose, as a component of the lignocellulose in the plant cell wall, constitutes the most abundant plant biomass on land and is a highly homogeneous polymer made up of glucose, a 6-carbon sugar, linked together in a chain of β-1,4-glycosidic bonds (**Figure 1**). The polymer strands are arranged in parallel to form cellulose microfibrils, with interstrand hydrogen bonding leading to a highly stable crystalline lattice (5). Thus, although cellulose is of the same sugar composition (i.e., glucose chains) as the first-generation bioenergy feedstock starch, its crystallinity renders cellulose highly recalcitrant to enzymatic degradation. However, the extent of cellulose degradation can be enhanced by increasing its amorphous regions.

Hemicellulose

Hemicellulose is the second most abundant plant polymer on our planet and exhibits an amorphous structure. Compared to cellulose, it is a more heterogeneous polymer. Different sugar units such as glucose, xylose, mannose, galactose, arabinose, and fucose are found in various hemicelluloses (6). However, xylans represent the most abundant hemicellulosic polymers and are made up of the 5-carbon sugar xylose joined in β-1,4-glycosidic bonds as the backbone (**Figure 1**), with varying side chain decorations depending on the source of the plant tissue. Xylans thus may be classified as (*a*) homoxylans (consisting of xylose linked in β-1,4- and β-1,3-glycosidic bonds) and (*b*) arabinoxylans, consisting of xylose linked together in β-1,4-glycosidic bonds as the backbone, with arabinose residues linked to some xylose residues at the *O-2* and *O-3* positions. The arabinose residues may also be linked to phenolic compounds, such as ferulic acid or *para*-coumaric acid (7), that in turn may be linked to lignin through ester bonds, especially in grasses. (*c*) Glucuronoxylans are found in hardwoods and woody plants and consist of a xylose backbone with 4-*O*-methyl

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glucuronic acid residues linked to the *O-2* position of xylose. (*d*) Arabinoglucuronoxylans are potentially an important feedstock for bioenergy production, because they represent the typical xylan found in grasses, such as those targeted in the biomass-to-biofuel industry, and have a xylose backbone with side chains of arabinose and methyl-glucuronic acid (MeGA) (6). A hemicellulose that is also of bioenergy interest is mannan, a heterogeneous polysaccharide found in softwoods and classified as homomannan, galactomannan, glucomannan, and galactoglucomannan (8).

Pectin

Although found in most bioenergy feedstocks, pectin is often not considered in discussions concerning optimization of the use of plant cell wall for bioenergy production. Pectins, however, represent the most complex of the plant cell wall polysaccharides (9). In the cell walls of grasses, pectins are relatively low in proportion, i.e., 2–10%; however, their complexity and intimate association with the other plant cell wall components suggest that they deserve attention if extensive hydrolysis of bioenergy feedstocks is to be achieved. Pectins constitute a family of galacturonic acid–rich polysaccharides (**Figure 1**), including homogalacturonan, rhamnogalacturonan II, and xylogalacturonan (9). Homogalacturonan (a linear polymer of α-1,4-linked galacturonic acid) may be acetylated or methylated. The rhamnogalacturonans are more complex in structure, with a main chain of galacturonic acid, interspaced with rhamnose branched with chains of arabinans, galactans, and arabinogalactans. Thus, other sugars found inserted in the pectin polymer include ribose, galactose, arabinose, and sucrose (10). Importantly, pectins are responsible for adhesion of neighboring cells (11), and this has implications for the recalcitrance of bioenergy feedstocks to deconstruction.

Lignin

Lignin, unlike cellulose and hemicellulose, is a noncarbohydrate heterogeneous polymer that derives from the oxidative coupling of three different phenylpropane building blocks or monolignols. The lignin monolignols are *para*-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol, with their radicals represented as 4-hydroxybenzyl (H), guaiacyl (G), and syringyl (S), respectively (12, 13). The lignin monomers polymerize through coupling of two radicals, thus forming a dimer by β-O-4 aryl ether bonds. The complex, highly hydrophobic, branched polymer arises through polymerization of monomeric radicals with dimers, trimers, and oligomers (12, 14) (**Figure 1**). Depending on the plant type, the components of lignin monomers vary, with lignin from grasses made up of H, G, and S units; lignin from hardwood constituted of approximately equal amounts of G and S units; and lignin in softwood made up mostly of G units. Lignin is recalcitrant to decomposition, although organisms such as the white rot fungi *Phanerochaete chrysosporium*, *Ceriporiopsis subvermispora*, and *Trametes versicolor* elaborate diverse extracellular oxidative enzymes that synergistically and effectively degrade lignin (12). Lignin associates with plant cell wall polysaccharides through esterification with hydrocinnamic acids, such as *p*-coumaric acid and ferulic acid in grasses and other graminaceous plants (15). Sun et al. (15) reported that approximately 56–82% of *p*-coumaric acid is esterified to cell wall components (mainly lignin), and approximately half of the cell wall ferulic acid is etherified through its oxygen to the cell wall lignin components, with the rest esterified to cell wall hemicellulose and/or lignin in different plant materials. These interactions of lignin with plant cell wall polysaccharides strongly impede enzyme-based deconstruction of lignocellulose. Furthermore, lignin binds to enzymes during lignocellulose degradation, preventing enzyme recycling, and therefore additionally encumbering effective depolymerization of plant cell wall polysaccharides (16).

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SECOND-GENERATION BIOENERGY FEEDSTOCKS

In the narrative above, we discuss the provenance of the sugars targeted for second-generation biofuel production; as a follow-up, we examine some of the feedstocks targeted in the field as sources of lignocellulose. Significant among these feedstocks are the perennial grasses *Miscanthus* and switchgrass. Perennial plants, such as sugarcane (bagasse), energy cane, elephant grass, switchgrass, and *Miscanthus*, which use C4 photosynthesis, have high light, water, and nitrogen use efficiency compared to C3 plants, such as poplar. Thus, C3 plants can be a focus of regions with abundant rainfall, whereas the C4 plants can serve as the source of feedstocks in regions where rainfall is limited (17). In addition, corn stover and crop residues are excellent candidates for bioconversion to liquid fuels. There is also evidence that short-rotation coppices of woody crops, such as poplar and willow, can add to the range of bioenergy feedstocks. Also, importantly, in semiarid regions, the highly water-efficient agave and opuntia, both of which are crassulacean acid metabolism (CAM) plants, can serve as feedstocks (17, 18). To introduce the reader to these feedstocks and their potential to contribute to the biomass-to-bioenergy efforts, we provide a brief overview of some of these materials, especially their polysaccharide and lignin contents, two factors that impact their potential as sources for fermentable sugars.

Miscanthus

The giant grass *Miscanthus*, a fast-growing perennial grass, is now grown on most continents. The significant attention paid to this grass as a bioenergy feedstock is due in part to its reported high yield, its minimal requirement for nutrients and water, and the observation that it grows in locations where it will not compete with current food crops (19). Turner et al. (20) recently reported that *Miscanthus giganteus* contains approximately 45.5% cellulose, 29.2% hemicellulose, and 23.8% lignin, with six other cultivars also containing high quantities of these polysaccharides (cellulose 41.1–46.0%, hemicellulose 24.3–32.6%, and lignin 21.4–24.9%). Thus, in total, more than 70% of this giant grass could be available for bioconversion to liquid fuels.

Switchgrass

Panicum virgatum L. is a perennial grass and one of the predominant species found in tall grass prairies of North America. Its uses include restoration of native range (21) and as a forage (3). Through the efforts of the US Department of Energy (22), switchgrass has attained the status of one of the main herbaceous biofuel crops. Among its desired characteristics are its high yield, perennial nature, capacity to establish from seed, nativity to the United States, and adaptability to poor soils (22). Whereas agricultural residues such as corn stover are reported to be composed of 33–35% cellulose, 21–24% hemicellulose, and 17–22% lignin, switchgrass is reported to contain higher cellulose (38–40%) and lower lignin (15–19%) content (23). Wang et al. (24) have also reported switchgrass used in their studies as 39.4% cellulose, 20.3% hemicellulose, and 21.2% lignin, confirming this high cellulose content. However, analyses of switchgrass composition have tended to vary, ranging at cellulose of 32–45% wt, hemicellulose 21–31% wt, and lignin 12–28% wt (25–28). These differences have been attributed to several factors, including species and cultivar differences (22, 24).

Sugarcane Bagasse

In countries such as Brazil, where sugarcane production is carried out on a very large scale, the major target for biofuel production is the juice or sucrose content. However, the massive amount of bagasse produced from this process represents a significant source of sugars for further conversion

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to second-generation bioenergy. Lima et al. (29) have reported sugarcane bagasse from Brazil as containing 39.5% glucan and 28.5% hemicellulose. Thus, the capacity to convert the massive amounts of sugars present in sugarcane bagasse can be a game changer in terms of the economics of bioconversion of sugarcane to biofuels.

Agave

This very promising biomass source is a perennial produced commercially for fiber (30). However, large areas are planted to harvest for tequila production, using the stem, which contains large amounts of soluble carbohydrates, with the leaves burned or composted. Agave leaves are, however, unique in having a low lignin content (5–20%) compared to other lignocellulosic feedstocks (9–32%). Furthermore, several varieties of agave are reported to have low crystalline cellulose contents and high amorphous cellulose. These properties suggest that sugar release from agave leaves is likely to be higher than in traditional lignocellulosic feedstocks, an observation confirmed by Li et al. (31). This group reported that non-pretreated agave releases approximately 4–7 times more sugars than poplar wood and approximately 3–5 times more sugars than switchgrass. Importantly, however, some agave species, such as *Agave lurida* and *Agave nizandensis*, are considered endangered (32). Therefore, such species are not expected to be targeted as bioenergy feedstocks.

Opuntia

This feedstock is native to the Americas. However, opuntia is currently grown in countries across the globe, including Algeria, Sicily in Italy, Argentina, Brazil, Chile, Israel, Mexico, and South Africa. Importantly, opuntia also grows in semiarid regions of the world. Its fruits are consumed by humans; however, it can also be fermented into an alcoholic beverage and used as animal fodder. As a potential biomass for bioenergy production, opuntia contains even less lignin compared to agave and also has lower crystalline cellulose and higher amorphous cellulose compared with agave (33). This suggests that opuntia could be even more easily degradable by enzymatic means than agave.

THE RUMEN AND PLANT CELL WALL–DEGRADING MICROORGANISMS

Ruminants are unique in having a four-chambered stomach. The first compartment of the ruminant stomach, known as the rumen, is a large chamber inhabited by diverse microbes with representatives from each of the three domains of life, i.e., Bacteria, Archaea, and Eukarya (34). The bacterial domain is represented by organisms such as *Prevotella* and *Ruminococcus*. The diversity of the archaea in the rumen, on the other hand, is not completely understood but includes the hydrogenotrophic methanogens, such as *Methanobrevibacter ruminantium*, as well as the methylotrophic methanogens, such as the *Methanomassiliicoccus* spp., or the MMC group. The eukaryotes are represented by a large diversity of protozoa (including *Entodinium* spp. and *Isotricha* spp.) and anaerobic fungi (including *Neocallimastix* spp. and *Caecomyces* spp.). The rumen also harbors viruses and bacteriophages; however, knowledge on this group remains limited (35). The microbial population, which has evolved together with the ruminant animal, plays a critical role in the life of the host through a symbiotic relationship. Here, the ruminant animal provides to its symbiotic microbes critical conditions for growth, including a suitable temperature for growth (36–41°C); an oxygen-free environment; and most importantly, a carbon and energy source (masticated feed). The microbes in return degrade the masticated feed, using specialized enzymes encoded on their genomes, and importantly provide the host with energy in the form of short-chain fatty acids (acetate, propionate, and butyrate). The plant cell wall–degrading enzymes, evolved by the

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rumen organisms in this symbiotic relationship, thus represent a rich source of enzymes that can be harnessed to release fermentable sugars present in second-generation bioenergy feedstocks, the limiting step in the process of converting biomass to biofuels. Below, we present types of rumen microorganisms and their enzymes of lignocellulose degradation.

Bacteria and Their Classification Based on Substrate Utilization

Based on numbers, bacteria represent the dominant group of microbes in the rumen. Their numbers are estimated in the range of 10^{10}–10^{11} cells per gram of rumen contents (36). The rumen bacteria can be classified based on the range of substrates they utilize or ferment, which also provides insights to the enzymes elaborated by a given bacterium. Thus, rumen bacteria may be classified as specialists or generalists. The specialist rumen bacterium degrades and uses a single substrate, whereas a generalist degrades and uses multiple substrates (37). Examples of specialist rumen bacteria are *Fibrobacter succinogenes* and *Prevotella ruminicola*, which degrade and ferment cellulose and hemicellulose, respectively (38, 39). Those of generalists include *Butyrivibrio fibrisolvens*, which degrades and utilizes hemicellulose, starch, and pectin (40, 41), and *Ruminococcus albus*, known to degrade and utilize cellulose and hemicellulose (42, 43). Based on these observations, it can be assumed that the generalist rumen bacteria are equipped with a broader battery of enzymes that target diverse polysaccharides of the plant cell wall, whereas the specialists likely harbor enzymes specially evolved to degrade unique components of lignocellulose.

Bacteria and Their Classification Based on Enzymatic Action

Rumen bacteria have also been classified based on their known enzymatic action, and the authenticity of this classification has been confirmed in analyses of the genomes of keystone bacteria from the rumen ecosystem (39, 44, 45). Thus, bacteria such as strains of *F. succinogenes*, *Ruminococcus flavefaciens*, and *R. albus* are designated ruminal cellulolytic bacteria, indicating that they are equipped with the enzymes that enable cellulose degradation. Sequencing of the genomes of these bacteria demonstrated that although each is known phenotypically to degrade cellulose, the genomic contents are different. Thus, whereas the plant cell wall–degrading gene repertoire of *F. succinogenes* mostly targets cellulose degradation, the genomes of both *R. flavefaciens* and *R. albus* show organisms highly equipped with genes for the degradation of both cellulose and hemicellulose (39, 45, 46). These observations explain why *F. succinogenes* has a lifestyle of a specialist and the *Ruminococcus* spp. are generalists. Also, Morrison and coworkers (44), through genome sequencing, showed that the genomes of two rumen *Prevotella*, i.e., *P. ruminicola* and *Prevotella bryantii*, encode more than 100 enzymes mostly targeting hemicellulose degradation, therefore confirming the two organisms as hemicellulolytic bacteria. Thus, the lifestyles of the rumen bacteria elucidate their potential as sources of enzymes for depolymerization of different components of bioenergy feedstocks.

The Rumen Anaerobic Fungi, Their Classification, and Their Plant Cell Wall–Degrading Enzymes

The anaerobic fungi represent one of the two groups of eukaryotes in the rumen and were originally thought to be flagellated protozoa; however, they were later shown to be related to the aquatic fungal group Chytridiomycetes (47). These organisms are strictly anaerobic and represent one of the first colonizers of fibrous materials in the rumen. Since Orpin (48) isolated and characterized *Neocallimastix frontalis*, several other members have been isolated, cultured, and described, including *Piromyces*, *Caecomyces*, *Cyllamyces*, *Orpinomyces*, *Pecoramyces*, *Agriosomyces*, *Joblinomyces*, and *Feramyces* (49, 50). The rumen fungi are classified as either monocentric or polycentric based on their mode of reproduction. They are monocentric if only one region of the fungus has

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reproductive function and polycentric if multiple regions of the fungal body have reproductive function. Despite their low numbers in the rumen (up to 10^6 zoospores/g rumen contents), they are thought to be important contributors to plant cell wall degradation, as they secrete a wide range of polysaccharide-degrading enzymes, including those that target cellulose, hemicellulose, and pectin (51–53). Sequencing of the genomes of the anaerobic fungi has been challenging; however, they likely represent a treasure trove for discovery of new plant cell wall–degrading enzymes, as the available data point to organisms encoding a large repertoire of such enzymes (54, 55). Enzymes encoded by these fungi include the critical carbohydrate esterases (CEs) that cleave both the polysaccharide from its linkage to lignin moieties and the side chain decorations that impose steric hindrance on the enzymes that degrade the polysaccharide backbone (36, 56). Such enzymes are, therefore, important for efficient release of fermentable sugars.

The Rumen Protozoa, Their Classification, and Their Plant Cell Wall–Degrading Enzymes

The other eukaryotic organisms present in the rumen are the protozoa. These organisms are 10–100× larger than the bacteria and are easily discernible under low magnification during examination of rumen fluid by microscopy. The rumen protozoa are grouped into two types, the Entodiniomorphida and the Holotricha (57). Although they are known to engulf starch and therefore likely possess amylases for hydrolysis to glucose, evidence for possession of enzymes such as cellulases and hemicellulases is confounded by their predatory behavior in the rumen. Protozoa engulf large amounts of both rumen bacteria (57) and fungi (58), and the cellulolytic and hemicellulolytic enzymes associated with the protozoa are thought to derive from their prey. Growing axenic cultures of protozoa has been a big challenge to rumen microbiologists, leading to difficulty in accessing complete protozoal genomes to obtain a clear view of their plant cell wall–degrading enzymes. Thus, the capacity of the rumen protozoa to contribute enzymes of plant cell wall hydrolysis remains to be fully understood, especially because a recent report suggests they produce enzymes that target cellulose, xylan, and mannan for degradation (58, 59).

ENZYMATIC STRATEGIES FOUND IN THE FIBER-DEGRADING MICROORGANISMS IN THE RUMEN

Analyses of the enzymes deployed by the fiber-degrading microorganisms found in the rumen suggest that these organisms use different strategies, including one where enzymes are expressed and secreted to function as freely diffusing or freestanding enzymes and another where enzymes are expressed and incorporated into large molecular complexes known as cellulosomes (38, 60, 61). Based on current data, it is unclear which strategy dominates in the rumen, as it appears that the genomes of members of the anaerobic fungi (60) and bacteria such as the *Ruminococcus* spp. (45, 61) deploy cellulosomes, and other fiber degraders, such as *F. succinogenes* (39) and two *Prevotella* spp., utilize freestanding enzymes (44). Below, we briefly describe the freestanding and cellulosomal strategies.

Freestanding Enzymes

In this strategy, the microbes secrete multiple enzymes that target cleavage of the different linkages in the plant cell wall polysaccharides. The enzymes often act synergistically in cleaving the linkages in the polysaccharide and thus releasing the component sugars for metabolism (62). A major plant cell wall polysaccharide-degrading bacterium in the rumen, i.e., *F. succinogenes* S85, uses this strategy for cellulose degradation (62). The strategy makes sugars available to the community in a process known as cross-feeding. In a slightly different strategy, the *Bacteroidota* in the rumen,

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i.e., the genus *Prevetolla* and their relatives, encode cell surface–attached endo-acting enzymes that initiate polysaccharide degradation. This initial degradation leads to oligosaccharides that are then transported into the periplasm or cytoplasm, where other enzymes further degrade the products to their unit sugars for metabolism by the bacterium (63, 64). The latter strategy has been referred to as the selfish mechanism of polysaccharide degradation (65), in contrast to the communal or cross-feeding mechanism. The freestanding enzyme system is found in aerobic, anaerobic, and facultative anaerobic organisms and is therefore widespread among microorganisms.

The Cellulosome, a Multienzyme Complex

The cellulosome was discovered in the bacterium *Clostridium thermocellum* and is found in anaerobic organisms as a multicomponent, multienzyme complex tethered to the surface of the microorganism (66). It has been proposed as an enzymatic system evolved to enhance synergistic activity among the different enzymes expressed by the bacterium to degrade complex polysaccharides (67). The cellulosomal structures include surface layer homology modules, carbohydrate-binding modules (CBMs), scaffoldins, cohesins, and dockerins. The organizational strategy is usually based on the scaffoldin, a large noncatalytic polypeptide anchored to the bacterial cell surface through the surface layer homology or a sortase-mediated mechanism (68). Within the scaffoldins are noncatalytic structures known as cohesins, to which diverse enzymes are docked. Enzymes are incorporated into the cohesins through noncovalent interactions with peptides known as dockerins, encoded together with the various enzymes that cleave the target polysaccharides. Thus, each enzyme when expressed by the bacterium can be assembled onto or incorporated into the scaffoldin through interaction between the enzyme-linked dockerin and the cognate cohesin. Further complexities have been achieved through adaptor scaffoldins that can connect other scaffoldins. In addition to the major rumen bacterium *R. flavefaciens* and related bacteria (69), rumen anaerobic fungal species of the genera *Anaeromyces*, *Neocallimastix*, and *Piromyces* have been reported to encode cellulosomes (54). It has been reported that dockerins from a different rumen anaerobic fungus, e.g., the genus *Piromyces*, bind to scaffoldins from other anaerobic fungal genera, e.g., *Anaeromyces* and *Neocallimastix*. This finding suggests that the scaffoldin system is conserved across the rumen fungal phylum (54). The fungal cellulosomes in the rumen could, therefore, be a composite of enzymes elaborated by different anaerobic fungi (54). In contrast, Bayer et al. (67), using extensive mutational analysis, have reported species specificity in bacterial cellulosomes and further identified amino acid residues critical in different dockerins for interaction with their cognate scaffoldins. Importantly, the cellulosomal structure is reminiscent of a molecular LEGO® brick (70) and therefore allows construction of designer cellulosomes (71) for degradation of lignocellulose. The foregoing, therefore, suggests the fungal scaffoldin/dockerin pairing as a more amenable chassis for construction of designer cellulosomes.

ENZYMES FOR DEPOLYMERIZATION OF BIOENERGY FEEDSTOCK

As discussed above, the cell walls of terrestrial plants harbor a rich source of energy in the form of polysaccharides, i.e., cellulose, hemicellulose, and pectin. These polysaccharides are rich sources of sugars for bioconversion to second-generation biofuels. The sugars, however, exist in a maze of complex linkages that require cleavage before they can be fermented to biofuels. Below, we discuss the enzymes evolved in nature to depolymerize these polysaccharides. Furthermore, under each enzyme, we examine functional homologs that have been reported in rumen microbes and later discuss how they may be harnessed in the biomass-to-biofuel industry. Importantly, carbohydrate-degrading enzymes and other related enzymes are classified and curated at the CAZy database (**http://www.cazy.org**).

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The Cellulolytic Enzymes

Different microorganisms have evolved enzymes collectively known as cellulases to degrade cellulose. We discuss these enzymes below.

Endoglucanases. Endoglucanases or endo-1,4-β-glucanases are critical for initiation and extensive degradation of cellulose. These enzymes function via random cleavage of the β-1,4-glycosidic bonds found in cellulose, an activity that disrupts regions of the cellulose structure. The action of these enzymes leads to multiple end products of glucose chains (cello-oligosaccharides), and these serve as substrates for downstream cellulose-degrading enzymes. Hence, for extensive hydrolysis of cellulose, endoglucanases are indispensable and widespread, being found and biochemically characterized from organisms living across a wide temperature and pH range (72–74). Examples of these enzymes have been described in multiple glycoside hydrolase (GH) families, including GH families 5–9, 12, 44, 45, 48, 51, 61, and 74. The catalytic reactions of these enzymes use both the retention and inverting mechanisms (75, 76). Significant numbers of endoglucanases have been found and characterized from rumen bacteria, including *F. succinogenes*, in which more than 10 highly upregulated GH enzymes belonging to GH5, GH8, GH9, and GH48 were found to release end products that include glucose, cellobiose, cellotriose, cellotetraose, and cellopentaose. Enzymes of this organism were found to be mostly nonmodular, existing as single catalytic modules (S. Yoshida, R.I. Mackie & I. Cann, unpublished data). The genome of *R. albus* 8 also encodes multiple endoglucanases, with many of the characterized enzymes derived from GH5 modules linked to CBM4–9, CBM2, CB37, or CBM65 and in a few cases to a dockerin peptide. Thus, unlike *F. succinogenes*, the endoglucanases of this rumen bacterium are modular polypeptides. The end products of these endoglucanases include glucose, cellobiose, and cellotriose (77), a range that differs from that of the *F. succinogenes* enzymes. In *R. flavefaciens*, which encodes a cellulosome, its cellulase enzymes are reported to be dominated by multiple GH5 and GH9 polypeptides (14 and 12, respectively) (46). Furthermore, the rumen fungus *Orpinomyces* sp. strain C1A was reported to deploy endoglucanases of diverse families, including GH5, GH8, GH9, and GH45. In this latter report, phylogenetic analyses showed that many of the enzymes are closely related to counterparts from rumen bacterial genera, including the *Ruminococcus* (55). Similarly, *Orpinomyces joyonii celA*, an endoglucanase-encoding gene, has been shown to be horizontally transferred from the rumen bacterium *F. succinogenes* (78). These observations are significant and, coupled with the general lack of introns in rumen fungal GH genes, provide evidence for horizontal gene transfer of GH genes, and perhaps the cellulosomal system (discussed above), from the bacterial domain to the eukaryotic domain in the rumen. Acquisition of these genes then likely allowed effective fungal colonization of the rumen (78). Importantly, in all the above rumen organisms, data from the genomes and transcriptomes suggest that they deploy multiple and diverse endoglucanases that target cellulose for degradation.

Cellobiohydrolases. The enzyme cellobiohydrolase (CBH), also designated β-1,4-exoglucanase, is a critical component of the enzymatic activities required for cellulose depolymerization. The CBHs function by binding to the products or cleavage sites generated by the endoglucanases or the free ends in cellulose microfibrils and then processively slide along the chain of glucose, cleaving off cellobiose (2 glucose units). The enzymatic activities of CBH, therefore, should be highly enhanced in the presence of an endoglucanase, as the random cleavage activity of the latter amplifies the CBH binding sites. The active sites of CBHs are tunnel-like in nature and therefore facilitate threading of the cellulose chain as cellobiose units are cleaved off. The substrate binding tunnel may vary in length, and this may impact processivity, with the shorter versions being less processive (79). CBH activities have been reported mainly in GH families 5–7, 9, and 48, and the

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enzymes are often bimodular polypeptides through fusion of the GH module with a CBM (80, 81). Some of the CBMs, including the one fused to the CBH I of *Trichoderma harzianum*, are reported to enhance cellulose degradation when expressed separately and added to reaction mixtures containing cellulolytic enzymes (82). Thus, the CBMs are potential additives for enhancing cellulose degradation. The reactions catalyzed by these enzymes result in net retention or inversion at the anomeric carbon atom. Furthermore, whereas some CBHs carry out hydrolysis from the nonreducing end, others cleave substrates from the reducing end (83). Polypeptides with CBH activity have been found and characterized in the rumen bacterium *R. albus* 8 (84), and despite some evidence (based on polypeptide sequence analyses) that anaerobic rumen fungi, such as *Piromyces* sp. E2 (85) and the highly cellulolytic ruminal bacterium *F. succinogenes* (86), encode CBH enzymes, the supporting data are incomplete. In contrast, there is evidence that the CelA protein of *Neocallimastix patriciarum* is a CBH (87). In *R. albus* 8, three CBH enzymes with a GH9 catalytic module have been reported. Each catalytic module is fused to a CBM4–9 at the N-terminal region, and two of the polypeptides are further linked at the C-terminal region to a CBM37. As expected, cellobiose was the end product of hydrolysis of these enzymes (84).

β-Glucosidases and cellodextrinases. For most bacteria, the component of cellulose that enters the glycolytic pathway for metabolism is glucose; however, in some cases, such as organisms with a cellobiose phosphorylase, the components will be glucose and a phosphorylated glucose (α-glucose-1-phosphate) (88). Therefore, the cleavage of the end products of the combined action of endoglucanases and CBH is critical. Cellobiose dominates the end products of endoglucanase and CBH activities, and the β-glucosidase enzyme splits the disaccharide into glucose units. Cellobiose accumulation inhibits both endoglucanase and CBH activities; thus, β-glucosidase activity can be considered rate limiting in cellulose hydrolysis (89, 90). Members of this group of enzymes belong to the GH1 and GH3 families, and they hydrolyze their substrates with a net retention of configuration at the anomeric carbon atom (91). Importantly, some of these enzymes function as cellodextrinases by cleaving both cellobiose and longer chains of glucose (cellodextrins) to glucose units. Rumen bacteria such as *F. succinogenes* (92) and fungi such as *Orpinomyces* spp. (93) have been reported to produce β-glucosidases, and cellodextrinase activity has been reported in *F. succinogenes* (94), *B. fibrisolvens* (95), and *P. bryantii* (96). High glucose concentration inhibits the activities of many of these enzymes, and this has been attributed to glucose either blocking the active site or preventing the hydrolyzed product from leaving the active site (97). Surprisingly, although these enzymatic activities have been detected in rumen microorganisms, and they are critical for extensive hydrolysis of the ruminant diet, identification of their coding genes and definitive biochemical characterization of the gene products are limited in the literature.

Hemicellulolytic Enzymes

In the plant cell wall, cellulose microfibrils are intertwined with hemicellulose, a heterogeneous sugar complex. Thus, for extensive and efficient degradation of plant cell wall polysaccharides, enzymes that target degradation of the associated hemicelluloses are required. This fundamental requirement for plant cell wall degradation is portrayed even in specialist organisms such as *F. succinogenes* S85, which expresses some hemicellulolytic enzymes when grown on cellulose (S. Yoshida, R.I. Mackie & I. Cann, unpublished data). Below, we describe the enzymatic activities that degrade the most common hemicellulose in bioenergy feedstocks, yielding sugars including xylose and arabinose for fermentation.

Endoxylanases. The endoxylanases or endo-1,4-β-xylanases function similarly to the endoglucanases by cleaving their polysaccharide substrate in a random fashion. However, the predominant substrate in grasses is arabinoxylan (98). The enzymes, therefore, cleave the β-1,4-glycosidic

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linkages between the xylose residues, generating end products of different lengths, i.e., xylo-oligosaccharides (short chains of xylose) and xylo-oligosaccharides decorated with arabinose side chains and/or other moieties found in the polysaccharide. The endoxylanases are found in GH5, 7, 8, 10, 11, and 43; however, GH10 and GH11 appear to be the most well-characterized enzymes in rumen microbes, including *R. albus* (99), *P. bryantii* (38), and *P. ruminicola* (100, 101). There are differences in how these enzymes process the substrate, but studies that probe this mechanism are rare. For example, the GH10 and GH11 cleavage products from arabinoxylan have arabinose at subsite +1 and subsite +2, respectively (102). This observation, which suggests variations in end products, is consistent with information derived from three-dimensional structures of the enzymes in complex with oligosaccharide products (103, 104). The nature of the oligosaccharide products generated by these enzymes, therefore, will impact their subsequent degradation to fermentable sugars. In fact, these complex end products, with side chain decorations, have been suggested to restrict organisms that lack the appropriate enzymes from accessing these nutrients (105). Importantly, the rumen bacterium *P. bryantii* was demonstrated through transcriptomic analyses to deploy multiple endoxylanases, including those derived from GH5, GH43, and GH10 families, when growing on wheat arabinoxylan. Analysis of the polypeptide architectures showed that the endoxylanases were composed of either single catalytic modules or multiple modules that included CBM and/or esterase modules together with the catalytic modules (38). In *P. ruminicola*, this bacterium was also reported to encode multiple endoxylanases (101), and biochemical characterization of a bimodular polypeptide with an endoxylanase and ferulic acid esterase activities has been reported (100). Furthermore, in the *R. albus* 8 genome, we identified at least five endoxylanases, two of GH10 and three of GH11 catalytic modules. Except for one polypeptide, each endoxylanase was a multi-modular protein that harbored the catalytic module together with two of either CBM9, 22, or 37 and in two cases linkages to CE family 4 modules. Expression and biochemical characterization of these complex polypeptides clearly showed degradation of arabinoxylans, but with a clear difference in the end products of the GH10 and GH11 enzymes (99). Endoxylanase activities are also present in the rumen eukaryotes, with an esterase/xylanase reported in the anaerobic fungus *N. patriciarum* (106) and a GH11 endoxylanase reported in the rumen protozoan *Polyplastron multivesiculatum* (107).

β-Xylosidases. The β-xylosidases, also known as xylan-1,4-β-xylosidases, function by cleaving xylose residues from the reducing ends of xylans or arabinoxylans. The role of the β-xylosidases is comparable to that of cellodextrinases, which cleave glucose from cello-oligosaccharides. The random cleavage action of endoxylanases generates multiple binding and cleavage sites for β-xylosidases and thus facilitates extensive degradation of the xylose backbone, leading to depolymerization of the polysaccharide. Synergism in substrate depolymerization is, therefore, observed in reactions containing both an endoxylanase and a β-xylosidase. The β-xylosidases fall into five GH families: GH3, 39, 43, 52, and 54 (108); however, the most characterized enzymes are in the GH3 and GH43 families. Their reaction mechanisms lead to either the inversion (GH43) or retention (GH3, 39, 52, and 54) of the stereochemistry at the anomeric carbon (CAZy database). From rumen microbes, three β-xylosidases have been expressed from *P. bryantii* (GH3 enzymes, two with PA14 insertions in their C-terminal half) (96), with others from *P. ruminicola* (GH3 with a PA14 insertion in the C-terminal half) (100) and *R. albus* 8 (GH43 with C-terminal CBM13) (99). Although the rumen anaerobic fungi are known to elaborate this enzymatic activity (109), whether genes encoding members of this enzyme have been recombinantly expressed for biochemical characterization is unclear.

α-Arabinofuranosidases. The α-arabinofuranosidases are critical enzymes that display synergistic activity with other hemicellulose-degrading enzymes, such as endoxylanases and β-xylosidases,

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by cleaving off L-arabinose side chains present in the polysaccharides. The latter side chains can exert steric hindrance on the backbone-cleaving enzymes and therefore impede polysaccharide depolymerization. The α-arabinofuranosidases are exo-acting enzymes and cleave diverse bonds including α-1,2- and α-1,3-linkages to arabinofuranosides of arabinan, arabinoxylan, and arabinogalactan. In many of these cases, their action debranches the complex polysaccharide. These enzymes belong to GH3, GH43, GH51, GH54, and GH62. The GH43 arabinofuranosidases cleave terminal α-1,5-linked arabinofuranosides, whereas GH51 and GH54 cleave both α-1,2- and α-1,3-arabinofuranosyl moieties from arabinan and xylans (110). Importantly, the side chain structure may be more complex, because some of the arabinose may be esterified to ferulic acid. Members of this group of enzymes use catalytic mechanisms with either net retention or net inversion of the stereochemical configuration at the anomeric carbon (CAZy database). Rumen microorganisms known to extensively degrade hemicelluloses, including both the bacteria and the anaerobic fungi, have been shown to possess this activity. Some of the earliest enzymes with this activity cloned and expressed from a rumen organism include an enzyme from *P. ruminicola* (101) and the *B. fibrisolvens* xylB, a bifunctional arabinofuranosidase/β-xylosidase (111). More recently, a GH43 β-xylosidase/α-L-arabinofuranosidase and a GH30 α-L-arabinofuranosidase/β-xylosidase (112) have been reported from a metagenome, and Moon et al. (99) have also expressed and characterized an α-L-arabinofuranosidase from *R. albus* 8. Despite evidence that the rumen anaerobic fungi produce α-L-arabinofuranosidase (113), identification and full biochemical characterization of such rumen fungal enzymes are rare in the literature.

α-Glucuronidases. In addition to the arabinose side chains on the xylose backbones of xylans and arabinoxylans, some xylose residues may be linked to MeGA at the *O-2* position. These substitutions are important in hardwoods and softwoods but also in grasses and cereals (114). Removal of the side chain MeGA substitution is required to achieve extensive hydrolysis of glucuronoxylans. The enzymes that cleave MeGA are known as α-glucuronidases, and this cleavage activity may be restricted to the terminal nonreducing end of the xylo-oligosaccharides (GH67 family) (115), or as in the GH115 family, the cleavage may be more robust, i.e., at both termini and at internal positions (114, 116). Activity is thought to be restricted by MeGA proximity to acetylations (117). The enzymes belong to GH4, GH67, and GH115 families (118, 119), although activity for the GH4 has been tested mostly on artificial substrates. An inverting mechanism has been reported for the GH67 group (115). The importance of these enzymes is often underscored by their presence in bacterial gene clusters encoding xylanolytic enzymes (120). An earlier report on the presence of an α-glucuronidase in the rumen was in the culture of the anaerobic fungus *Piromonas communis* (now known as *Piromyces communis*) grown on birchwood xylan (121). Subsequently, an α-glucuronidase gene *agu67A* from *R. albus* was identified and expressed, and the activity of the recombinant protein reported (99). Furthermore, based on metagenomics, a rumen α-glucuronidase RUM630-AG exhibited synergistic activity with an endoxylanase (122).

Esterases. Hemicelluloses such as xylans, especially arabinoglucuronoxylans, are often acetylated at the *O-2* and *O-3* positions of the xylose chain. Furthermore, as noted above, other side chains in the form of ferulic or *p*-coumaric acid may be esterified to the 5′-OH of the arabinofuranosyl decorating the xylose backbone. The ferulic and *p*-coumaric acid may then ether-link or ester-link to lignin or dimerize with other ferulic or *p*-coumaric acid groups (123), increasing the complexity of the plant cell wall structure. Other hemicelluloses, such as mannans and glucomannans, and pectins may also be acetylated and ferulolylated. These highly intricate chemical linkages, therefore, present significant challenges to the enzymes that depolymerize plant cell wall polysaccharides. To overcome these limitations to accessibility of the sugars in the polysaccharide, many microbes have evolved in their genomes genes that encode enzymes that cleave these esterified

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decorations. These specialized enzymes are grouped as CEs. The CEs of importance to bioenergy feedstock depolymerization are the feruloyl esterases, *p*-coumaroyl esterases, and acetyl xylan esterases. The latter, also known as xylan deacetylases (124), often exhibit broad specificities in cleaving ester bonds. The acetyl xylan esterases fall under many CE families, including CE1–CE7, although some classifications based on sequence similarity have come under question (124). The feruloyl esterases, on the other hand, cleave off ferulic acid and other hydroxycinnamic acids (123) and are currently classified in the CAZy database under only the CE1 family. Many polypeptides exhibiting this activity are unassigned to any family, including a highly potent ferulic acid esterase discovered by our group (64). Although their activities were not considered in earlier enzyme cocktails for feedstock depolymerization, these auxiliary enzymes are critical for efficient hydrolysis of hemicelluloses and pectins and, for that matter, the plant cell wall polysaccharides. There are many reports on the presence of esterases in rumen microorganisms. However, many are based on metagenome-retrieved genes; therefore, which organisms they come from are often unknown (125–127). There are also several reports where the genes have been identified and expressed and the gene products have been demonstrated to exhibit esterase activities based on either artificial or natural substrates or both (56, 100, 128–131). Although reports on *p*-coumaroyl esterases from the rumen are rare, the activity has been purified from the fungus *Neocallimastix* strain MC-2 (132).

PECTIN-DEGRADING ENZYMES

Although low in lignocellulose contents, pectin is a very complex polysaccharide, and its interactions with hemicellulose and lignin are likely to impact enzyme accessibility to degradable polysaccharides. Due to its complexity, pectin degradation requires many enzymes, which can add to the cost of the substrate depolymerization step of the biomass-to-biofuel process. However, a few members of these enzymes could be enough to enhance sugar release from bioenergy feedstocks. We discuss below the enzymes required for depolymerization of this complex component of the plant cell wall.

Protopectinases

Protopectinases are reported to be the first enzymes necessary for pectin degradation, because they convert insoluble to soluble pectin, which can then be depolymerized by other enzymes that participate in pectin degradation (10).

Polygalacturonases

These enzymes function to depolymerize the polygalacturonic acid chain of pectin through cleavage of the α-1,4-glycosidic bonds linking the galacturonic acid units. Two types of these enzymes have been reported, i.e., the endo-polygalacturonases and the exo-polygalacturonases. The former cleave the backbone to generate oligo-galacturonic acids, with the latter cleaving the pectic acids to liberate mono-galacturonate (133).

Pectin Esterases

Due to the presence of methyl ester linkages in the galacturonan backbone of pectin, efficient degradation requires side-chain removal. The enzymes that perform the de-esterification, leading to pectic acid and methanol (10, 134), are pectin methyl hydrolases, pectin methylesterases, and pectase. Cleavage of the side chains exposes the galacturonic acid backbone for extensive depolymerization by the polygalacturonases.

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Arabinanases

Arabinans, although different from pectins, are often distributed widely among the hemicellulose and pectin components of the plant cell wall (135). The polysaccharide is composed of an α-1,5-linked L-arabinofuranosyl residue as the backbone, with substitutions of α-1,2- and/or α-1,3-linked L-arabinose side chains (110, 136). Arabinans increase pectin complexity and recalcitrance to depolymerization. Arabinan hydrolysis requires the endo-acting enzymes endo-1,5-α-arabinanases and side chain–removing α-L-arabinofuranosidases. The endo-acting enzymes, generally from the GH43 family, cleave the α-1,5 linkages of the backbone using an inverting catalytic mechanism. Synergistic degradation of the arabinan is then engendered through the exo-acting arabinofuranosidase, releasing arabinose monomers (110).

Galactan-Degrading Enzymes

Pectin components, e.g., rhamnogalacturonan-I regions, are often covalently associated with type I arabinogalactans, which are characterized by a β-1,4-galactan backbone. In contrast, type II arabinogalactans are characterized by β-1,3-galactan or β-1,6-galactan backbones (137). To degrade these polysaccharides, diverse enzymes are required, including an endo-β-1,4-galactanase, exo-β-1,4-galactanase, β-galactosidase, ferulic acid esterase, α-L-arabinofuranosidase, and endo-α-1,5-arabinanase (138). The functions of several of these enzymes have been described above. Hydrolysis of type II arabinogalactans is extremely complex, requiring many more enzymes, including an endo-β-1,3-galactanase, exo-β-1,3-galactanase, β-1,6-galactanase, endo-β-1,6-galactanase, β-galactosidase, α-L-arabinofuranosidase, β-glucuronidase, α-L-rhamnosidase, and β-L-arabinopyranosidase. Both rumen fungi (52) and bacteria (139) have been reported to produce pectin-degrading enzymes. For example, Deng et al. (140) reported two exo-polygalacturonases derived from a complementary DNA from rumen fluid. Both polypeptides belong to the GH28 family. Furthermore, Yuan et al. (141) retrieved several DNA fragments encoding polygalacturonases and pectate lyases from a sheep rumen. These genes were predicted to have come from the genera *Butyrivibrio*, *Prevotella*, and *Fibrobacter*, and Williams & Orpin (142) earlier reported galactan-degrading enzymes from three different anaerobic fungi. Thus, although the enzymes for pectin degradation have not been studied extensively in rumen microorganisms, published results suggest that such enzymes are elaborated by these organisms.

LIGNIN-DEGRADING ENZYMES

As noted above, the functions of lignin include protecting the plant from microbial enzyme degradation (12). Thus, lignin removal could improve biomass degradation. Here, we briefly discuss some of the enzymes of bacterial lignin degradation that can augment rumen enzymes in lignocellulose depolymerization.

Lignin-Degrading Peroxidases and Laccases

The lignin peroxidases generally degrade lignin with the help of H_2O_2 as a co-substrate that initiates the oxidation reaction through deprotonation. These enzymes are classified as manganese peroxidases, lignin peroxidases, dye-decolorizing peroxidases, and versatile peroxidases, and in bacteria they are found in members of the α-proteobacteria, γ-proteobacteria, and actinomycetes. The reader is referred to Abdel-Hamid et al. (12) for a detailed description of these interesting enzymes. The other group of lignin-degrading enzymes, laccases, belong to a superfamily of multicopper oxidases. They catalyze oxygen-mediated oxidation of lignin and other phenol-containing compounds. Importantly, laccase reaction leads to formation of water as a side product, and with

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no other cofactor requirement and freely available oxygen as a second substrate, using laccases for lignin degradation is considered advantageous (143). There are reports of bacteria isolated from the buffalo rumen harboring peroxidases and laccases for lignin degradation (144, 145). However, what roles these enzymes play in lignin degradation in the anaerobic environment of the rumen is uncertain. Importantly, as shown in **Figure 2**, there are polypeptides in rumen fungi and bacteria annotated as ligninases, and despite some evidence for the presence of oxidases that function without the need for mediators (146), the functions of these putative ligninases await comprehensive investigation. Despite the uncertainty in rumen microbial degradation of lignin, lignin-degrading enzymes from other sources clearly can be used to augment cocktails of polysaccharide-degrading enzymes aimed at releasing fermentable sugars from lignocellulose for fermentation to biofuels. Our reasoning here is based on the lignocellulose degradation being carried out under aerobic conditions.

A STRATEGY FOR PROSPECTING FOR NOVEL FUNCTIONAL POLYSACCHARIDE-DEGRADING ENZYMES FROM THE RUMEN

Functional metagenomic approaches have been used for prospecting for rumen bacterial enzymes with the capacity to degrade polysaccharides derived from the plant cell wall (146, 149). This approach describes the diversity and potential for plant cell wall degradation, but it is incomplete. Although previous reports demonstrate the feasibility of this approach (150, 151), we would like to recommend a three-phase approach, in which metagenomic sequencing, functional metagenomics, and metatranscriptomics (and also metaproteomics, based on availability of resources) are used in parallel for discovery of novel enzymes to augment enzyme cocktails for plant cell wall hydrolysis.

In this strategy, illustrated in **Figure 3**, samples can be collected from domesticated or wild ruminants (rumen) and other herbivores (cecum and colon), especially those on recalcitrant and poor-quality high-fiber diets. Representative samples are then spun at low speed or squeezed through layers of filter cloth to remove large particles and then immediately spun at high speed to separate and collect the liquid fraction and fiber-adherent microorganisms. Samples are then divided into three: The first sample is used for metatranscriptomics, to which RNAlater solution is added immediately and stored at −80°C until used for extraction of community RNA to establish which genes are being transcribed and are active in fiber hydrolysis. A second sample is used for community genomic DNA extraction for metagenomic sequencing using a long-read approach such as Pacific Biosciences or Oxford Nanopore Technologies sequencers to provide high-resolution species diversity and composition. Finally, a third sample is used for extraction of community genomic DNA for functional metagenomics analysis. Although this approach provides very useful information, we emphasize the importance of cultivation-based approaches to isolate the potent lignocellulose degraders in the samples collected. The reasoning here is that the microbes that have evolved to initiate extensive degradation of lignocellulose are likely to harbor the most potent enzymes. Furthermore, understanding their evolved strategies will be more informative in assembling the game-changing enzyme cocktails sought by the field.

POTENTIAL FOR SCALE-UP OF ENZYME COCKTAIL PRODUCTION

Enzyme mixtures, as biocatalysts to release fermentable sugars, represent a significant cost in biofuel production (152). Large-scale production of many commercial enzymes relies on filamentous fungi, as they have several advantages over bacterial-based systems (153). Members of the filamentous fungal genera *Aspergillus*, e.g., *Aspergillus niger*, and *Trichoderma*, e.g., *Trichoderma reesei*, are the workhorses in industrial enzyme production. The filamentous fungi, however, display very

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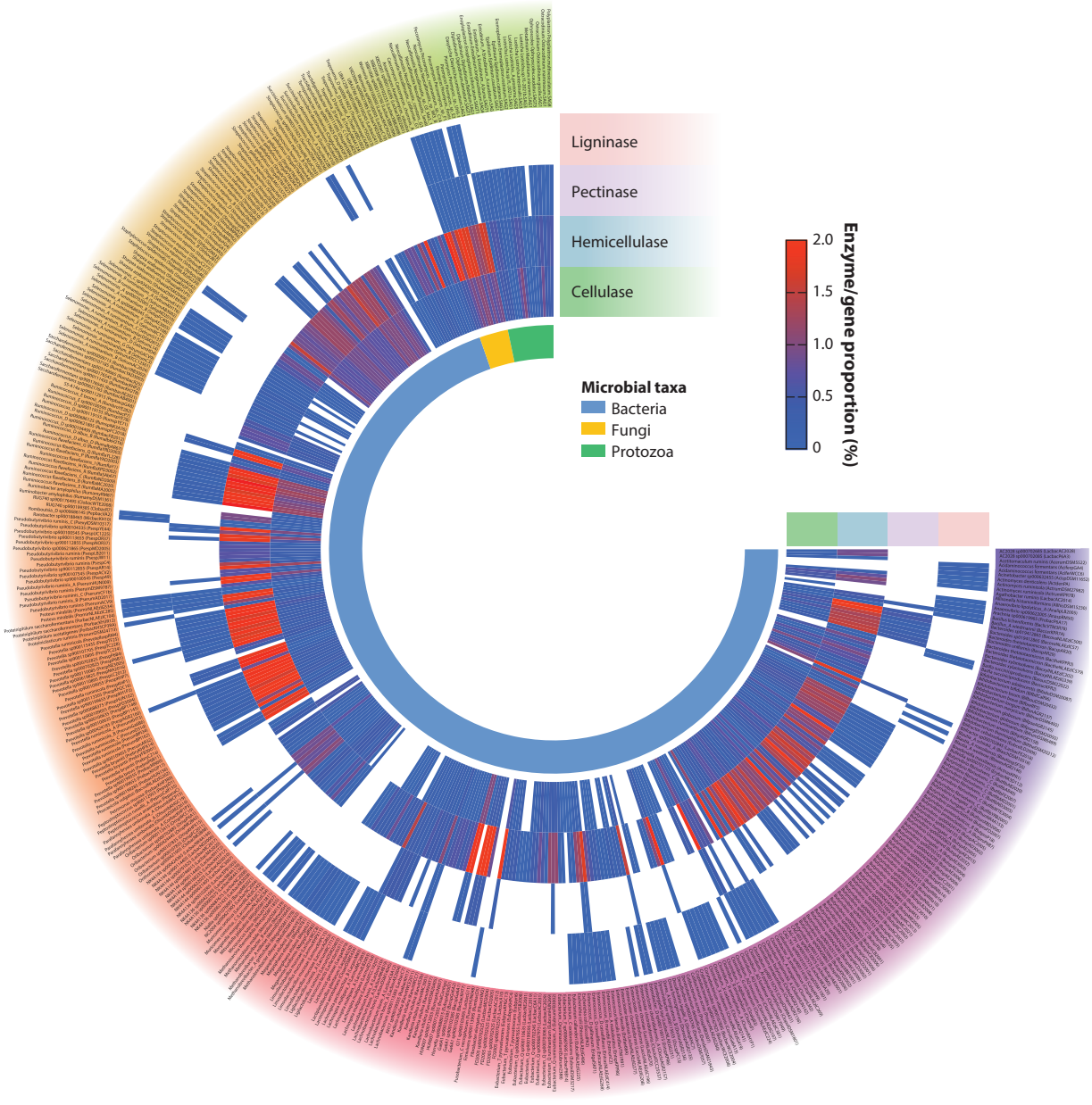

Figure 2

Enzymatic activities evolved by rumen microorganisms. In this ring diagram, the outermost ring represents the diversity of microorganisms, and the next four rings represent the proportions of genes encoding putative ligninases, pectinases, hemicellulases, and cellulases, respectively, in the genomes of the organisms. The innermost ring represents the clustering of the microbes analyzed under rumen bacteria (*blue*), fungi (*yellow*), and protozoa (*green*). The scale indicates the proportion of the genes based on the entire coding sequences in the genomes. The methanogens are not largely considered in this representation, as they are not primary contributors to plant cell wall degradation. The sources for the genomes are the Hungate1000 project (147), the JGI MycoCosm Fungal Portal (**https://jgi.doe.gov/our-science/science-programs/fungal-genomics/mycocosm-jgi-fungal-portal/**), and Li et al. (148).

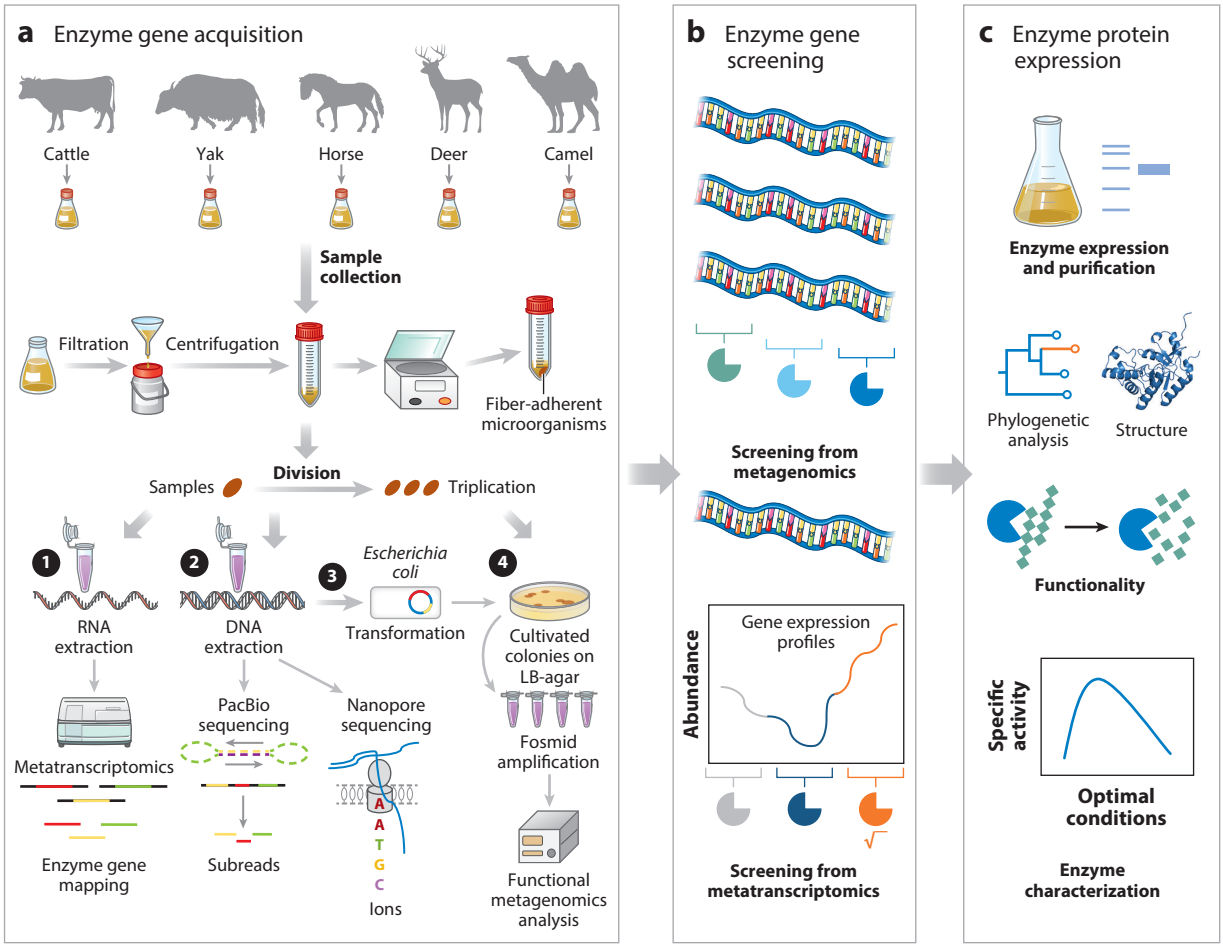

Figure 3

A proposed outline for prospecting for cell wall–degrading enzymes. (*a*) Samples are collected from domesticated or wild true ruminants (e.g., the rumen of cattle and yak) and other herbivores (e.g., cecum or colon of camel, deer, or horse) on recalcitrant and poor-quality high-fiber diets. Samples are immediately spun at low speed or filtered through layers of filter cloth. Next, the samples are centrifuged at high speed to collect the liquid phase and fiber-adherent microorganisms (precipitate). Collected samples are subjected to metatranscriptomics (RNA extraction), direct metagenomics (no library construction), and fosmid libraries for activity screening and also for direct insert sequencing. (*b*) Assays are developed for screening to discover activities of interest. (*c*) Genes of interest are expressed to confirm biochemical activities and phylogenetic analyses, and proteins are expressed in large quantities for three-dimensional structure determination.

complex life cycles (153), and our understanding of their physiology lags behind that of bacterial models, such as the Gram-negative *Escherichia coli* and the Gram-positive *Bacillus subtilis*. Despite this observation, the filamentous fungi have many advantages as vehicles for large-scale enzyme production, including the potential to produce proteins in large quantities and extracellular production (or secretion) of proteins, which eliminates a costly cell lysis step (153). Coupling these advantages to a better understanding of the physiology of the filamentous fungi, especially those harnessed for enzyme production, and a better community effort to share details of conditions that have resulted in enhanced yields of recombinant enzymes will help accelerate progress toward overcoming this bottleneck in lignocellulosic biofuel production.

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In a previous review (153), we outlined some areas in filamentous fungal physiology that need special attention in terms of enzyme production, including topics such as messenger RNA stability, protein quality control, endoplasmic reticulum–associated protein degradation, and extracellular proteolysis. Other considerations that may improve the practicality of enzymes for lignocellulose depolymerization include enzyme recycling, reduction of nonproductive lignin-mediated adsorption of enzymes (154), and enzyme stability during storage (152). In line with the above, we refer the reader to a recent report on *T. reesei* strain Br_TrR03. By applying bioprocess optimization, pilot plant scale-up, techno-economic analysis, and life cycle assessment for recombinant enzymes production, researchers increased protein productivity by twofold and cellulase activity (filter-paperase, or FPase) by 1.6-fold. The researchers also noted temperature and pH as the most significant factors that impacted enzyme production by this fungal strain; furthermore, their enzyme preparation lost only 10% of the original activity during storage for 1 year in glycerol at room temperature (25 ± 4.0°C). Also, importantly, in their techno-economic analysis, the authors reported an enzyme cost below what is targeted in the biofuel field (152). This group, therefore, presented an experimental design that yielded results demonstrating the promise associated with enzyme-based hydrolysis of lignocellulosic biomass. Compared to chemical treatments, enzyme-based hydrolysis of lignocellulose has many advantages; for instance, the degradation process could be controlled due to enzyme specificities; the best pH and temperature conditions are predetermined; and enzymes are proteinaceous and therefore biodegradable, making them less hazardous to the environment. It is anticipated that the field will adapt the approach outlined for strain Br_TrR03 to build on the reported success through production and implementation of highly efficient rumen-derived recombinant enzyme cocktails for lignocellulose hydrolysis.

FUTURE PERSPECTIVES

Understanding the mechanisms underlying how rumen bacteria degrade plant cell wall polysaccharides was the holy grail of rumen microbiologists from the earliest studies pioneered by Robert Hungate, the father of rumen microbiology (155). A molecular understanding of how these fascinating bacteria have evolved through millennia to access the growth substrates bound in lignocellulose, while making metabolizable energy in the form of volatile fatty acids available to the host ruminant, was expected to spur rumen manipulations to increase rumen function and animal productivity. Advancement in the field was, however, hampered by the lack of genomic information for these bacteria; therefore, rumen microbiologists from this earlier era were limited to biochemical characterization of single enzymes and sometimes just partial (not full-length) sequences of enzymes. Although the studies described in this review have demonstrated that rumen microbes elaborate all the enzymes required to break most of the linkages in lignocellulose, the former or earlier approach had no chance of elucidating the molecular mechanisms and enzyme complexity that underlie the depolymerization of the intricate substrate.

The capacity to sequence microbial genomes should have been a game changer for the above efforts, although the cost, in the millions of dollars, was initially daunting and limiting. Currently, a microbial genome can be sequenced for as low as $400, thus making it possible to achieve what was beyond earlier rumen microbiologists. However, instead of efforts being directed to sequencing complete genomes to allow research designed to probe the strategies evolved by the efficient fiber degraders, effort has now shifted to sequencing community genomic DNA to search, express, and biochemically characterize individual enzymes. To advance the field of plant cell wall degradation, a shift back to the aim of learning directly from the rumen bacteria how they achieve hydrolysis of lignocellulose is essential. For example, *F. succinogenes* S85 is known to grow rapidly on crystalline cellulose, but currently the precise enzymatic mechanism is unknown.

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Importantly, from the above analyses and current data, lignocellulose degradation clearly cannot be achieved cheaply, as effective degraders appear to employ a large arsenal of CAZymes, sometimes of the same family or of the same apparent activities (38, 156). Thus, instead of single or a few endoxylanases in an enzyme cocktail, multiple endoxylanases deployed by a hemicellulose degrader would be the correct approach. The successful approach to reconstituting an effective plant cell wall–degrading enzyme cocktail will be based on the combined complement of enzymes used by a rumen cellulose degrader, a hemicellulose degrader, and a pectin degrader (based on transcriptomic or proteomic analyses) to access energy from a lignocellulosic substrate. These enzymes can then be supplemented with lignin-degrading enzymes and other unique and potent enzymes (from the prospecting effort in **Figure 3**) that target rate-limiting steps in lignocellulose hydrolysis.

A better understanding of the activities and uniqueness of enzymes secreted by the rumen bacteria is also needed and should go beyond simple assays, such as thin-layer chromatography, to more sophisticated structural analysis offered by mass spectrometry. This will help decipher, for example, why a fiber degrader such as *R. albus* is wired with multiple GH10 and GH11 enzymes to degrade xylan. More importantly, a new cadre of rumen microbiologists/enzymologists, with strong carbohydrate chemistry analytical skills, is required to extend the work started by the pioneers of this field. These contributions should help improve rumen function and efficiency, while decreasing the yield of the greenhouse gas methane and also making available potent enzyme mixtures to spur the field of renewable biofuels. Both end results are critical to the sustainability of our planet.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENT

The authors are aware that there could be significant contributions in this field that have not been cited in this review. It is our hope, however, that the review will spur interest in the research area in general and therefore the research of all colleagues in the field. Research of I.C. and R.I.M. on plant cell wall–degrading enzymes was supported by the Energy Biosciences Institute and by the National Institute of General Medical Sciences of the National Institutes of Health under award no. RO1GM140306. The research of Y.C. and W.Z. on plant cell wall hydrolysis enzymes is funded by the National Science Foundation of China (32061143034, 32372905).

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